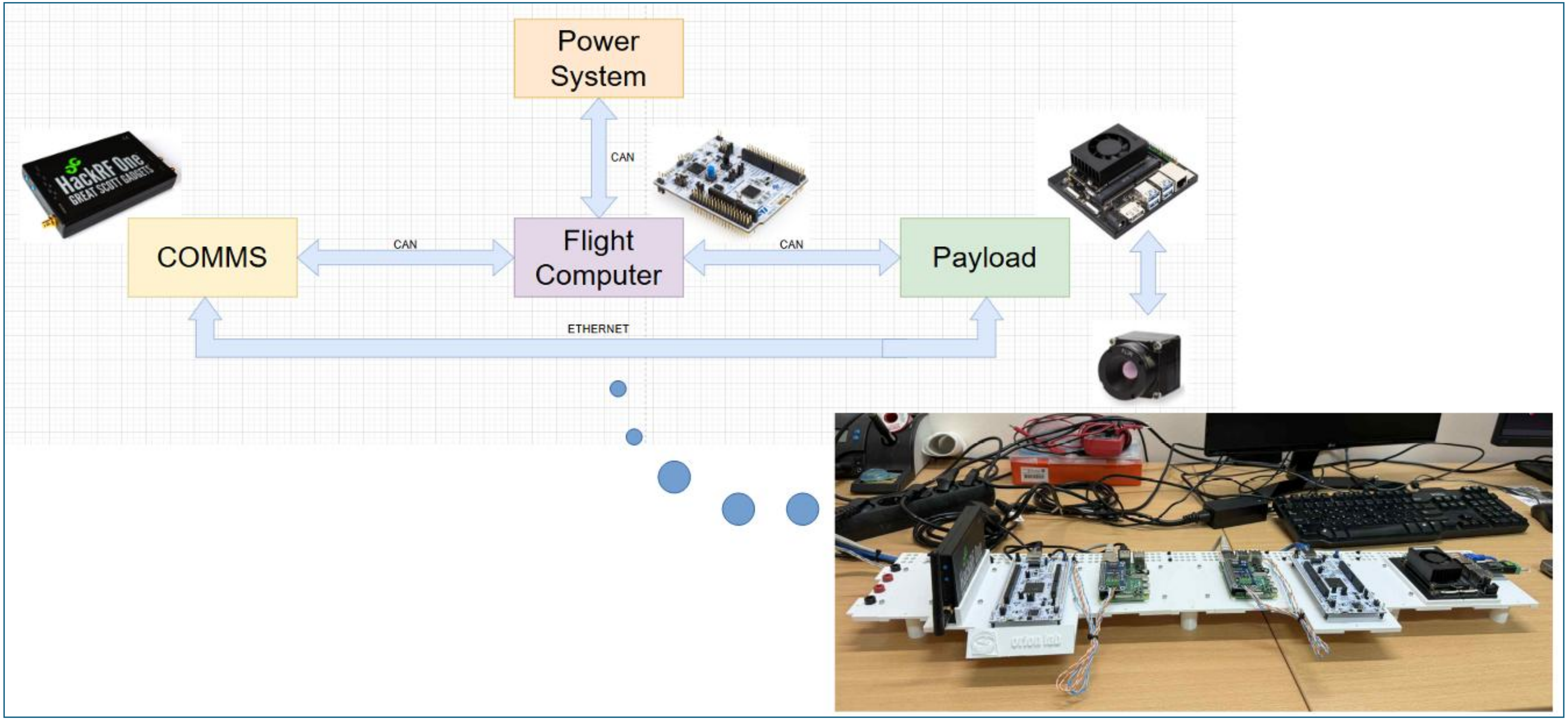
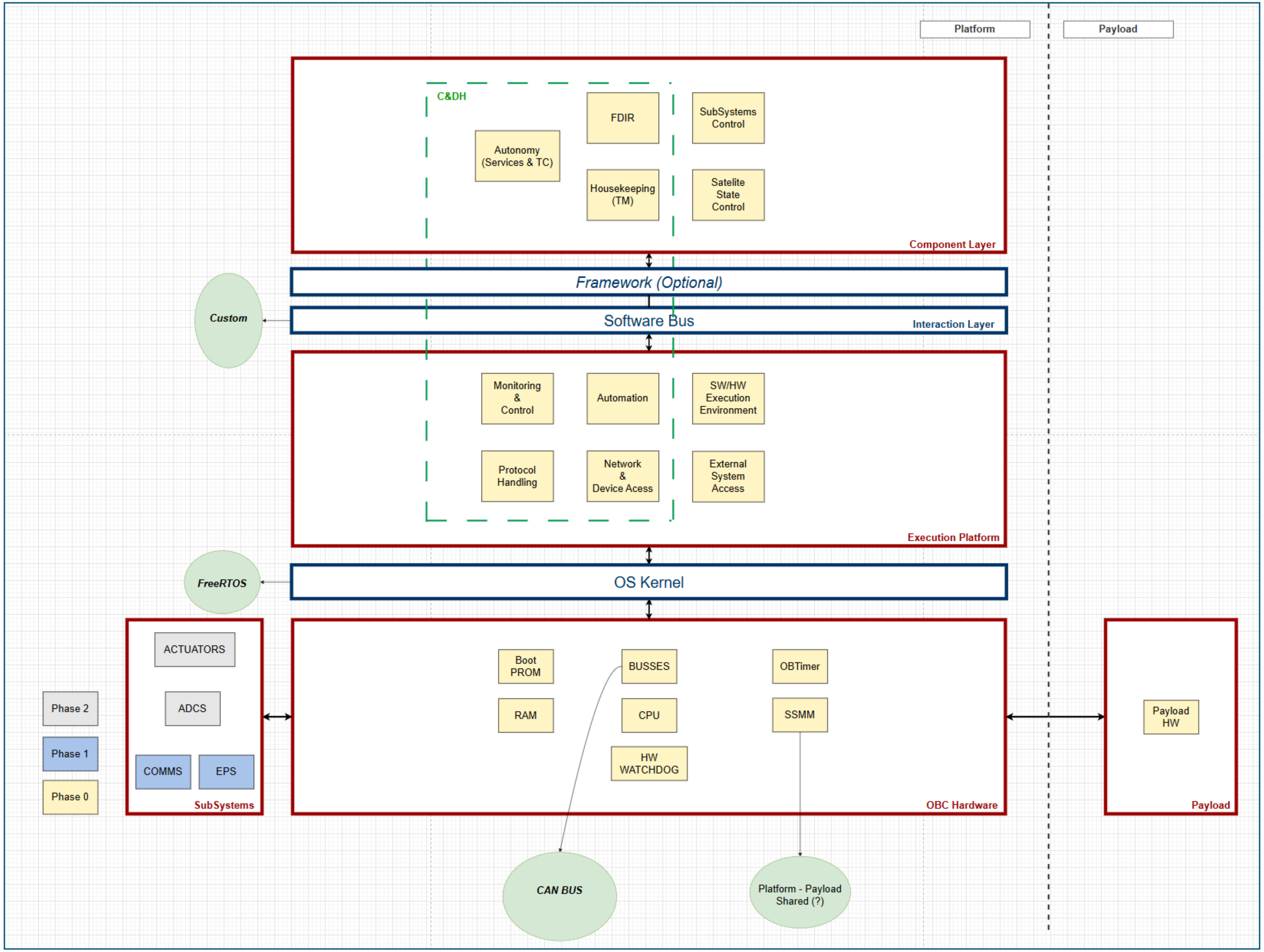
CubeSat Testbed – Architecture Overview



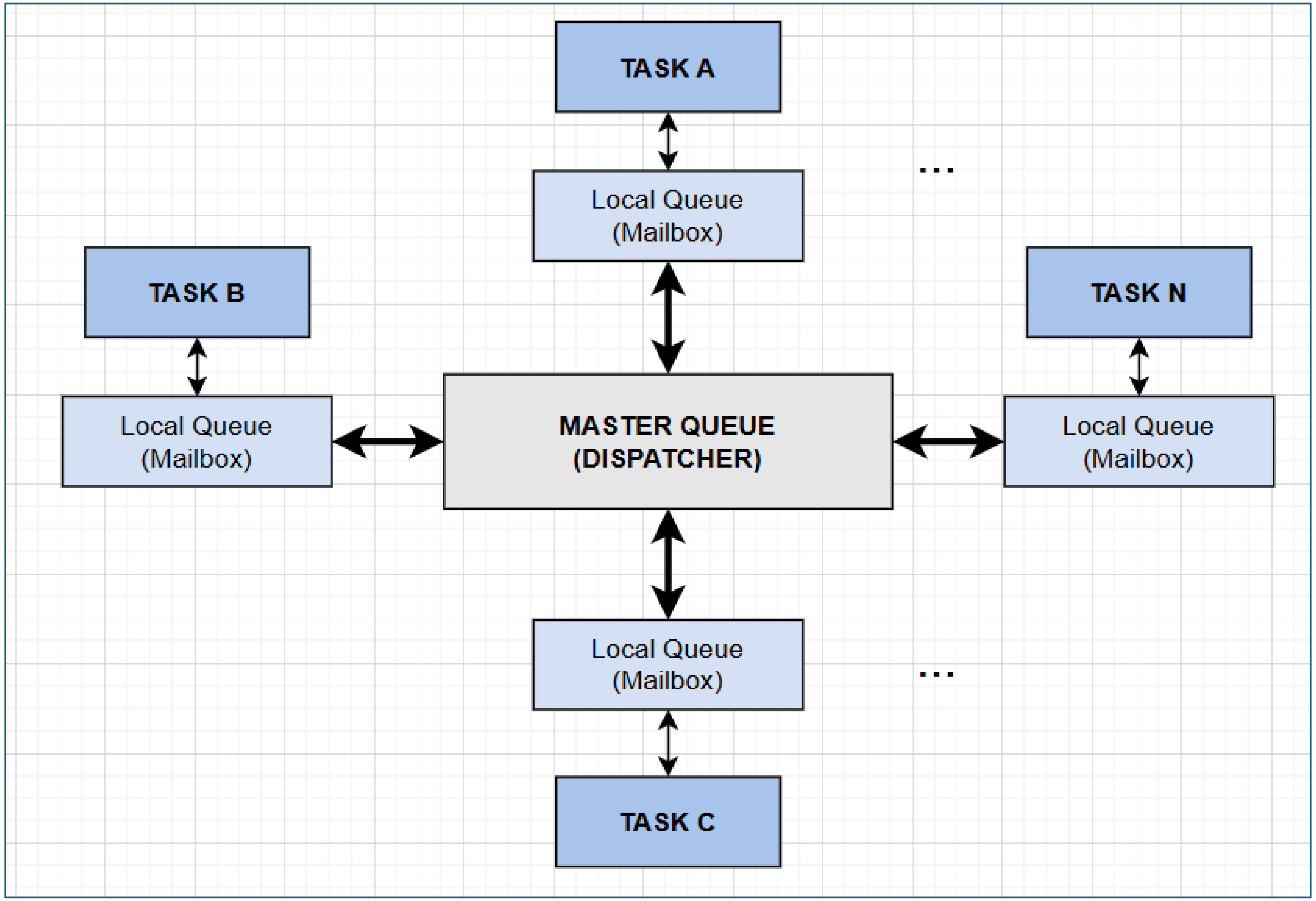
CubeSat Testbed – System Architecture



CubeSat Testbed – Software Bus Architecture

NOTE:
- Master Queue acts as a traffic controller
- Flow
  - Task A creates msg and pushes it to the master queue with TopicID
  - Task A blocks
  - Dispatcher wakes up since queue no longer empty
  - Dispatcher looks at the subscription table and pushes msg to subscribers' mailboxes
  - Subscribers wake up

NOTE:
- Need for multiple copies of the same msg when multiple tasks subscribe to the same topic
  - Zero-copy method!



CubeSat Testbed - Mission Requirements

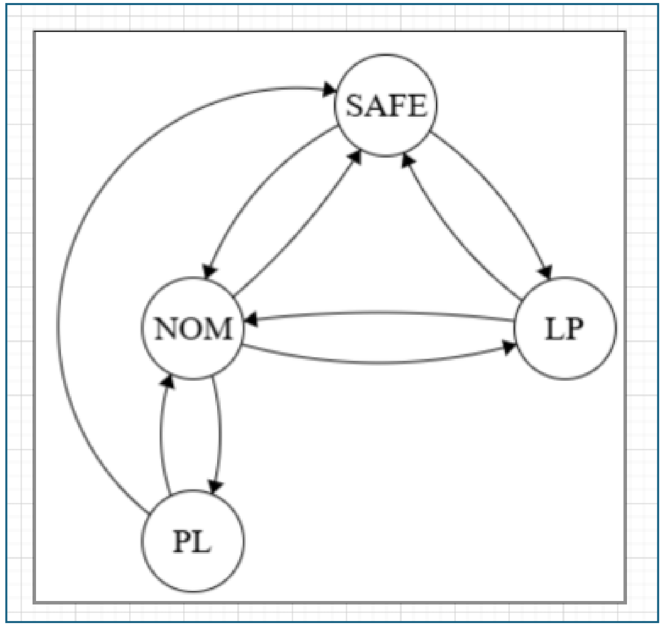
ID	Category	Description
MIS-OPS-01	Satellite Modes	The satellite must have modes to prioritize safety and achieve desired functionality
MIS-CTRL-01	Subsystems Control	The satellite shall provide centralized management of all onboard resources to ensure safe and correct functionality.
MIS-TM-01	Housekeeping & Telemetry	The satellite shall report subsystem health during every ground pass
MIS-TC-01	Ground Command Execution	The satellite shall react to commands received from the ground station
MIS-TS-01	Time Synchronization	The satellite shall maintain a global "Mission Elapsed Time" synchronized with Ground Station UTC.
MIS-REL-01	Reliability	The satellite shall recover from a total loss of ground-to-space communications.
MIS-REL-02		The satellite shall recover from critical software corruption to prevent permanent mission loss.
MIS-REL-03		The satellite shall recover from CPU hangs and software progress failures.
MIS-REL-06		The satellite shall maintain internal communication links even in the event of a primary bus failure.
MIS-REL-07		The satellite shall detect and recover from subsystem failure.
MIS-REL-08		The satellite shall ensure the integrity of critical system data against radiation-induced corruption.
MIS-PWR-01	Power Management	The satellite shall monitor its energy reserves to ensure safe operation.
MIS-PWR-02		The satellite shall monitor HW temps and ensure the system stays within operating range.
MIS-PL-01	Payload Management	
MIS-SUS-01	Sustainability	The satellite shall support In-Orbit Software Updates.

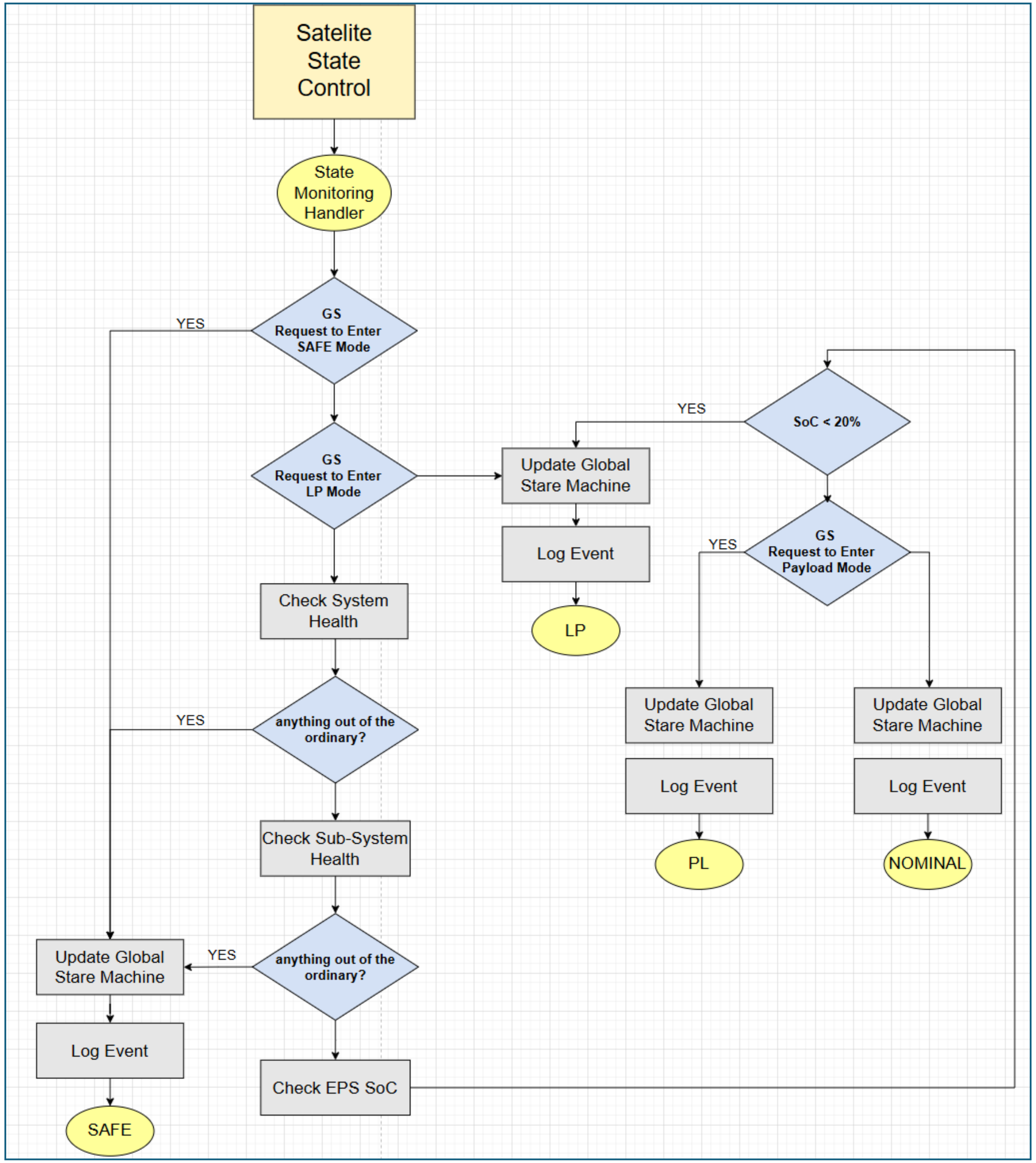
CubeSat Testbed – System Requirements

MIS-OPS-01 / Satellite Modes

MIS-OPS-01	Satellite Modes	The satellite must have modes to prioritize safety and achieve desired functionality

SYS-OBC-OPS-01	The OBC shall manage transitions between SAFE, NOMINAL, LOW POWER and SCIENCE modes based on current conditions and ground commands and inform the rest of the system.	MIS-OPS-01





CubeSat Testbed – System Requirements

MIS-CTRL-01 / Subsystems Control

MIS-CTRL-01	Subsystems Control	The satellite shall provide centralized management of all onboard resources to ensure safe and correct functionality.

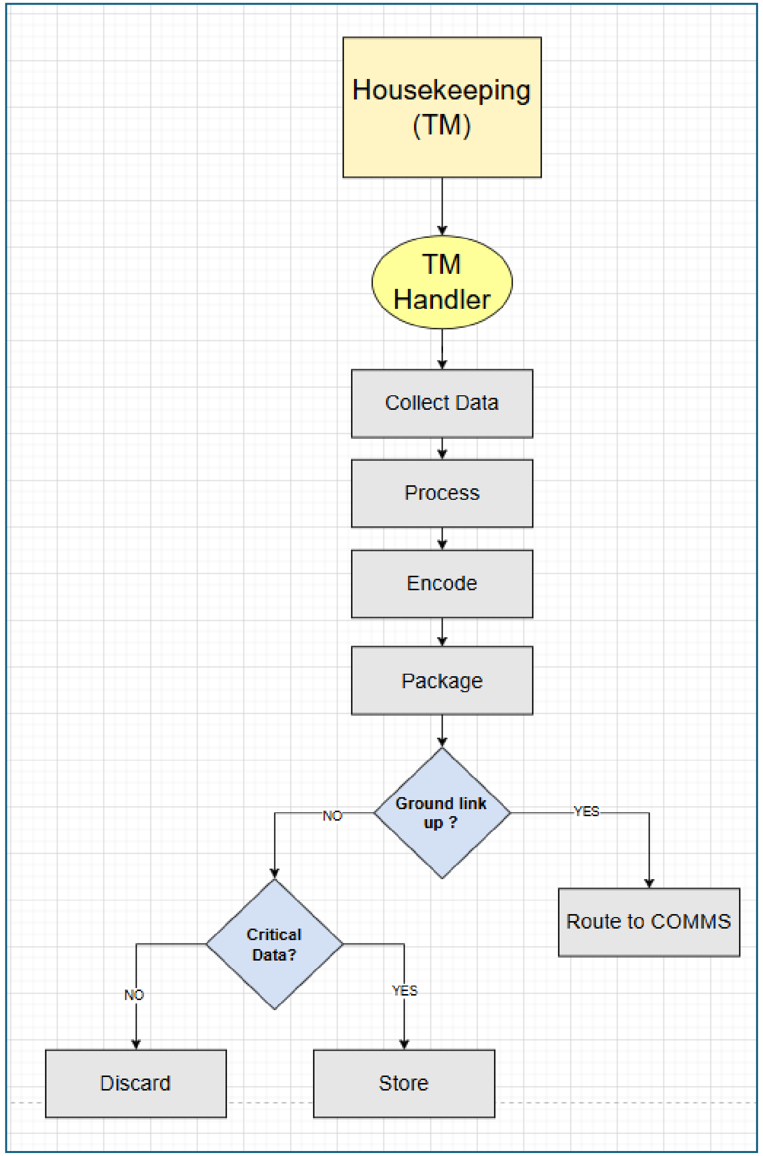
SYS-OBC-CTRL-01	The OBC shall serve as the bus master for all inter-subsystem communications	MIS-CTRL-01
SYS-OBC-CTRL-02	The OBC shall have the authority to control subsystems based on a TC or current conditions	

CubeSat Testbed – System Requirements

MIS-TM-01 / Housekeeping and Telemetry

MIS-TM-01	Housekeeping & Telemetry	The satellite shall report subsystem health during every ground pass

SYS-OBC-TM-01	The OBC shall collect housekeeping telemetry from all active subsystems including itself.	MIS-TM-01
SYS-OBC-TM-02	The OBC shall detect when the satellite is over GS and good comm has been established and initiate downlink	
SYS-OBC-TM-03	The OBC shall prioritize the downlink of "State of Health" packets over payload data during every GS pass.	

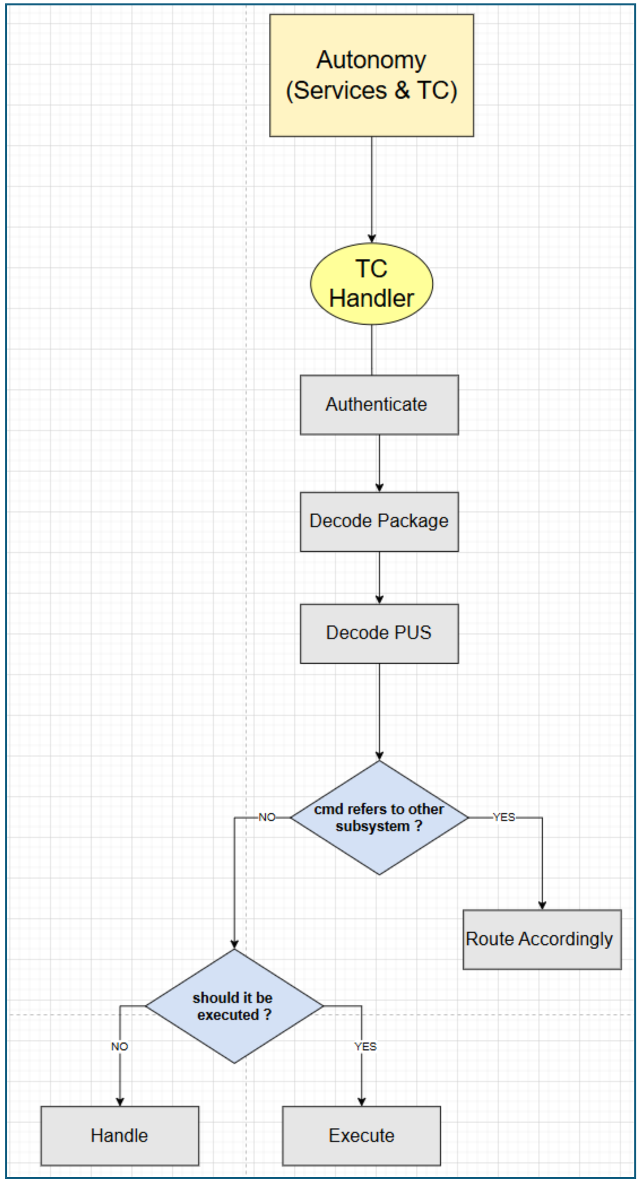


CubeSat Testbed – System Requirements

MIS-TC-01 / Ground Command Execution

MIS-TC-01	Ground Command Execution	The satellite shall be able to react to commands received from the ground station

SYS-OBC-TC-01	The OBC shall decode packets following the CCSDS standard.	MIS-TC-01
SYS-OBC-TC-02	The OBC shall authenticate every telecommand to prevent unauthorized access.	
SYS-OBC-TC-03	The OBC shall provide an ACK telemetry packet upon successful execution of a command.	
SYS-OBC-TC-04	The OBC shall reject and log any command that fails a Checksum (CRC) or format validation.	
SYS-OBC-TC-05	The OBC shall execute TC after successful authentication and verification	



CubeSat Testbed – System Requirements

MIS-TS-01 / Time Synchronization

MIS-TS-01	Time Synchronization	The satellite shall maintain a global "Mission Elapsed Time" synchronized with Ground Station UTC.

SYS-OBC-TS-01	The OBC shall act as the "Time Master" for the spacecraft, distributing a synchronization message to all subsystems.	MIS-TS-01
SYS-OBC-TS-02	The OBC shall allow the Ground Station to "Set" or "Adjust" the onboard clock via a TC.	

CubeSat Testbed – System Requirements

MIS-REL-01 / Reliability

MIS-REL-01	Reliability	The satellite shall be capable of autonomous recovery from a total loss of ground-to-space communications.

SYS-OBC-REL-01	The OBC shall implement a system timer that resets upon a GS kick and performs a system level Power-On Reset if it expires.	MIS-REL-01
SYS-OBC-REL-02	The OBC shall broadcast a periodic health "heartbeat" signal (Beacon) to allow the ground station to track the satellite and diagnose the communication status.	

CubeSat Testbed – System Requirements

MIS-REL-02 / Reliability

MIS-REL-02	Reliability	The satellite shall be recoverable from critical software corruption to prevent permanent mission loss.

SYS-ALL-REL-01	All subsystems shall recover from a corrupted main image by reverting to a "Golden" image on boot.	MIS-REL-02

CubeSat Testbed – System Requirements

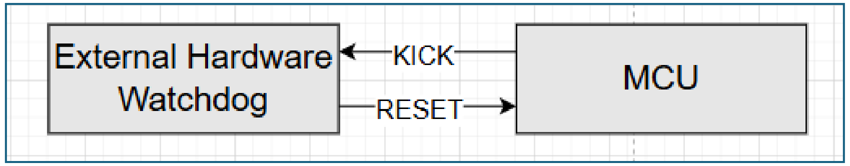
MIS-REL-03 / Processor Hangs

MIS-REL-03	Reliability	The satellite shall be capable of recovering from CPU hangs and software progress failures

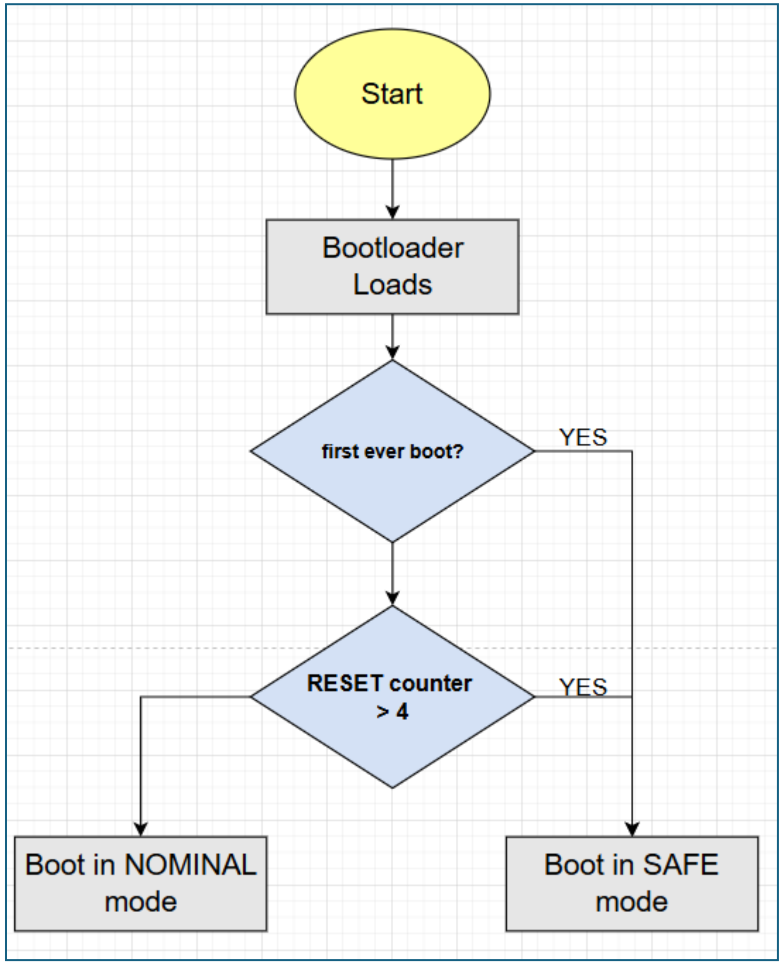
SYS-ALL-REL-02	All subsystems shall include an independent Hardware Watchdog capable of triggering a full CPU reset.	MIS-REL-03
SYS-ALL-REL-03	All subsystems shall maintain a "Persistent Boot Counter" in Non-Volatile Memory to track consecutive reset events and revert to Safe Mode after 3 consecutive resets occur.	
SYS-ALL-REL-04	All subsystems shall implement a Watchdog Manager Task that monitors the check-in status of all registered RTOS threads.	

CubeSat Testbed – System Requirements

SYS-ALL-REL-02

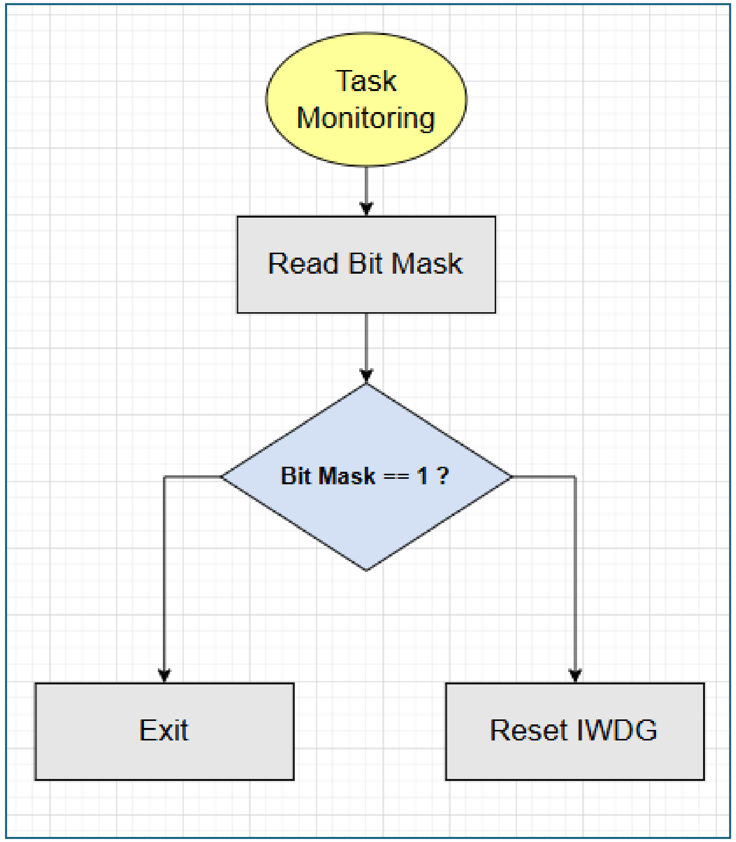


SYS-ALL-REL-03



CubeSat Testbed – System Requirements

SYS-ALL-REL-04

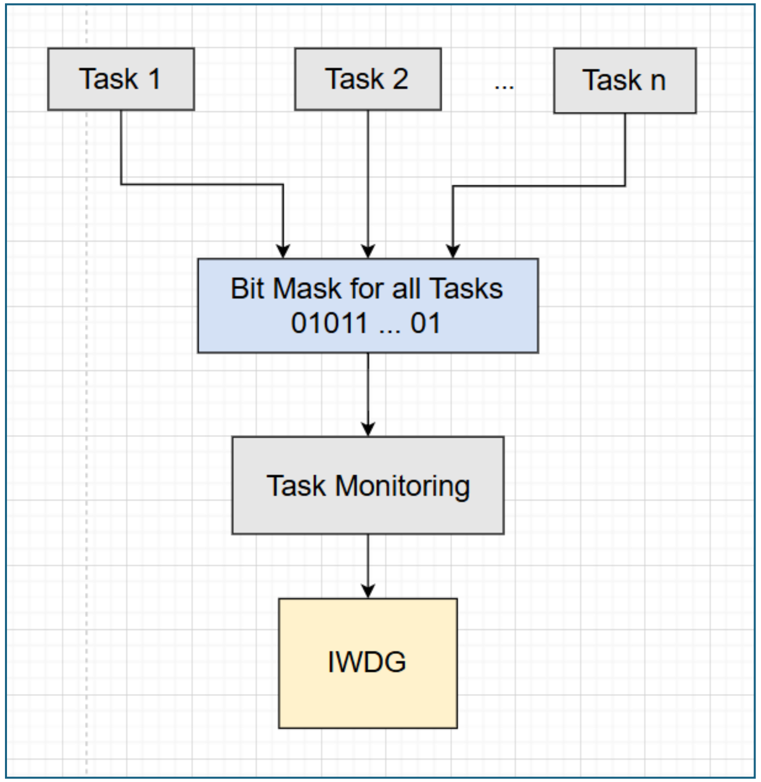


NOTE:

1. There should be a bit mask with a number of bits equal to the number of the tasks created. Each task should set its bit every time it completes an execution or write timestanp.

2. The Task Monitoring shall periodically check the mask's value. If it's 1 (or compare timestamps) then all tasks are making progress and the hardware WDG should be reset.

3. If x ms pass and the WDG has not been reset, a software reset will be generated by the WDG.



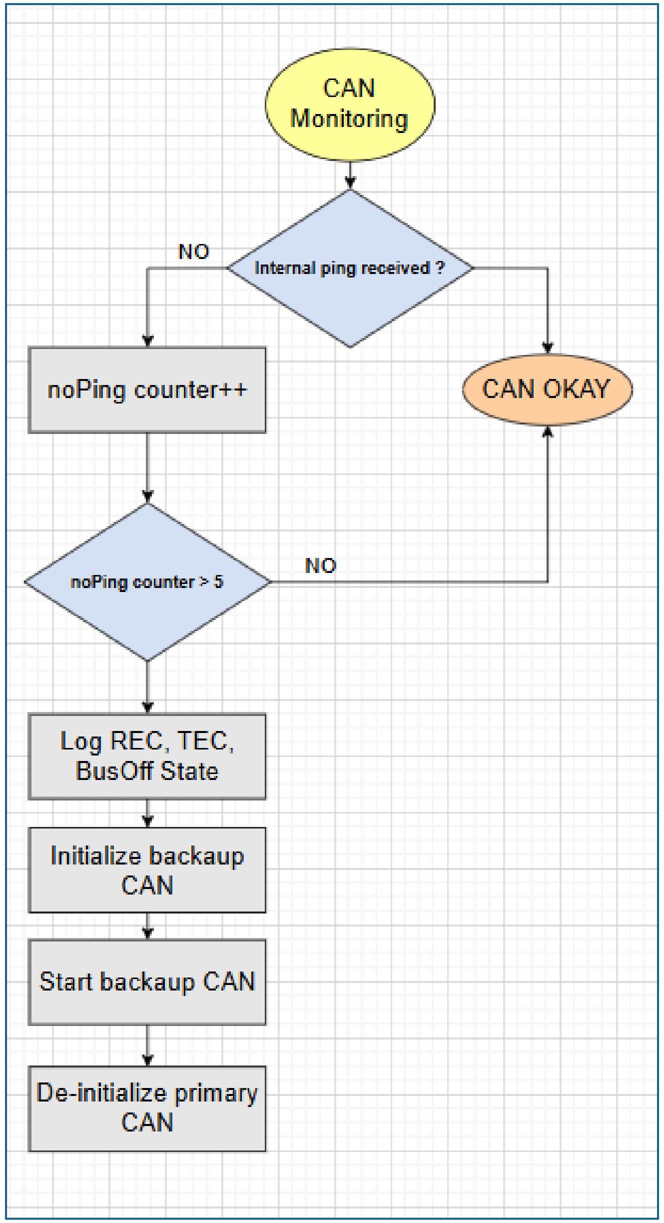
CubeSat Testbed – System Requirements

MIS-REL-06 / Reliability

MIS-REL-06	Reliability	The satellite shall maintain internal communication links even in the event of a primary bus failure.

SYS-OBC-REL-03	The OBC shall detect when intersystem communication has been compromised and revert to the back-up bus	MIS-REL-06

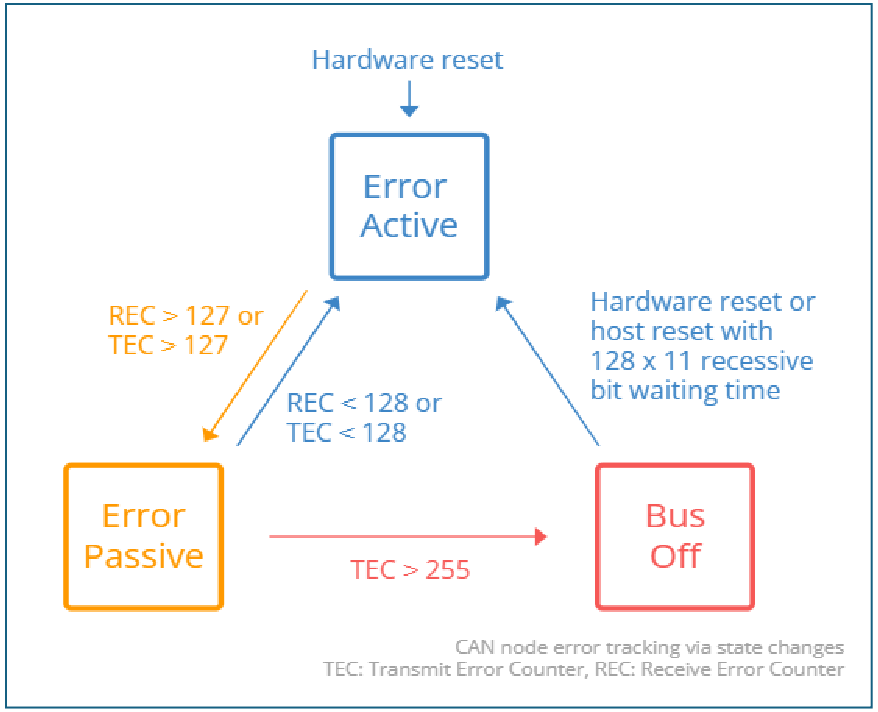
CubeSat Testbed – Flight Computer Functionalities



SYS-OBC-REL-03

NOTES:

The CAN Monitoring Task will periodically check for CAN activity. A separate task will transmit a ping back to the OBC. If the CAN Monitoring Task doesn't receive a ping for more than x5 consecutive times, then assume something is wrong and switch to the backup CAN.



https://www.csselectronics.com/pages/can-bus-errors-intro-tutorial

CubeSat Testbed – Flight Computer Functionalities

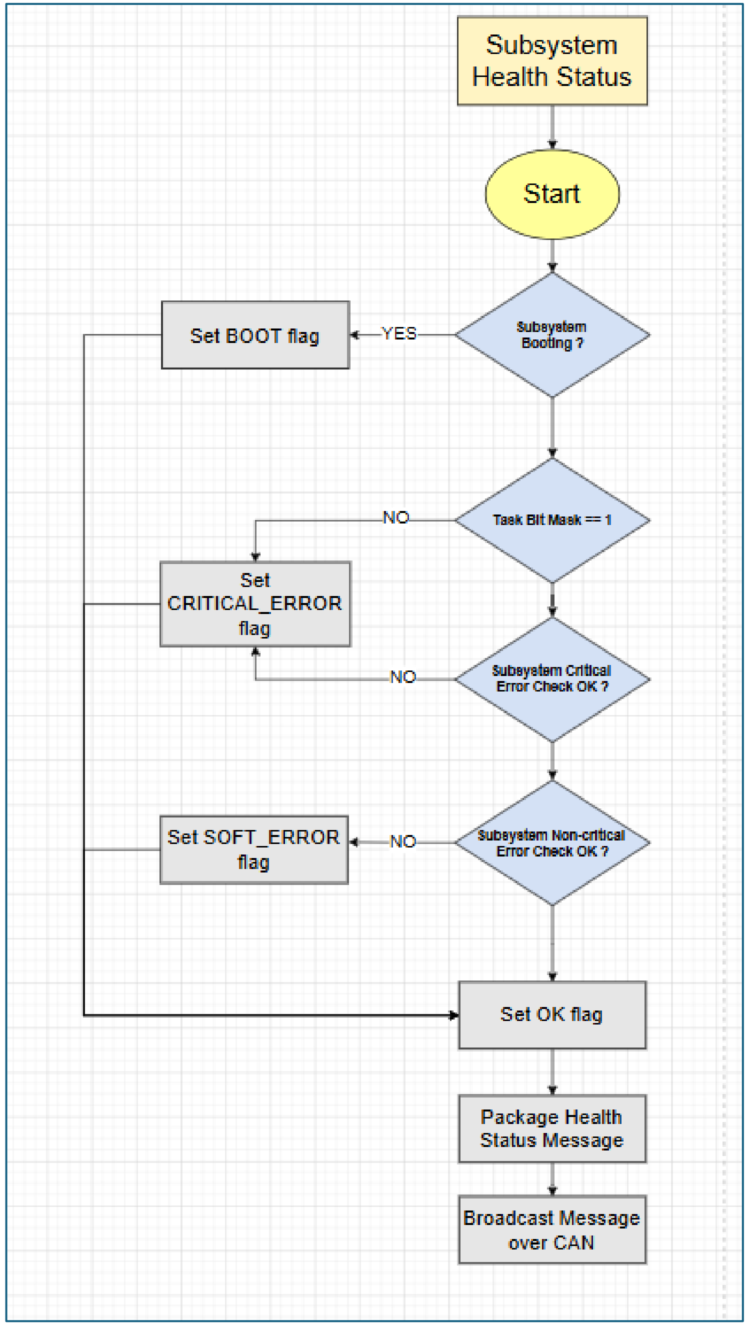
MIS-REL-07 / Reliability

MIS-REL-07	Reliability	The satellite shall detect and recover from subsystem failure.

SYS-ALL-REL-05	All subsystems must check internal health and periodically update OBC of their status.	MIS-REL-07
SYS-OBC-REL-04	OBC is responsible for checking HB health and escalate depending on the status.	

CubeSat Testbed – Flight Computer Functionalities

SYS-ALL-REL-05



CubeSat Testbed – Flight Computer Functionalities

SYS-OBC-REL-04